

Proposition: $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}^{\text{odd}}$ and $\exists k \in \mathbb{Z}$
Such that $n = m \cdot 2^k$

Base case: Let $n = 1$

$$\text{Then } n = 1 \cdot 2^0$$

Inductive Hypothesis: $\forall i \in \mathbb{Z} \ 1 \leq i \leq n$ such that
 $i = m_i \cdot 2^{k_i}$

Inductive Step: $n+1$

Case I: $n+1$ is odd

$$\text{Then } n+1 = (n+1) \cdot 2^0$$

Case II: $n+1$ is even

$$\text{Let } i = \frac{n+1}{2}$$

$$\text{Then } n+1 = 2i$$

$$= 2(m_i \cdot 2^{k_i})$$

$$= m_i \cdot 2^{k_i+1}$$

∴ any integer n can be written as the product of some odd integer and a power of 2, as was to be shown.

Q.E.D

Proof without Induction

Proposition: $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}^{\text{odd}}, 1 \leq m \leq n, \exists k \in \mathbb{Z}, 0 \leq k$

Such that

$$n = m \cdot 2^k$$

$$\frac{2j}{2} = j$$

Case I: n is odd

$$\text{Then } n = n \cdot 2^0$$

$$\frac{j}{2}, \frac{j}{4}, \frac{j}{8}, \frac{j}{16} \dots$$

Case II: n is even

Then $\exists j \in \mathbb{Z}, 1 \leq j < n$ such that $n = 2j$

I need to show that

① j is odd

or

② repeated division by 2 ends with an odd j

I can show it terminates, but why is an odd quotient guaranteed?